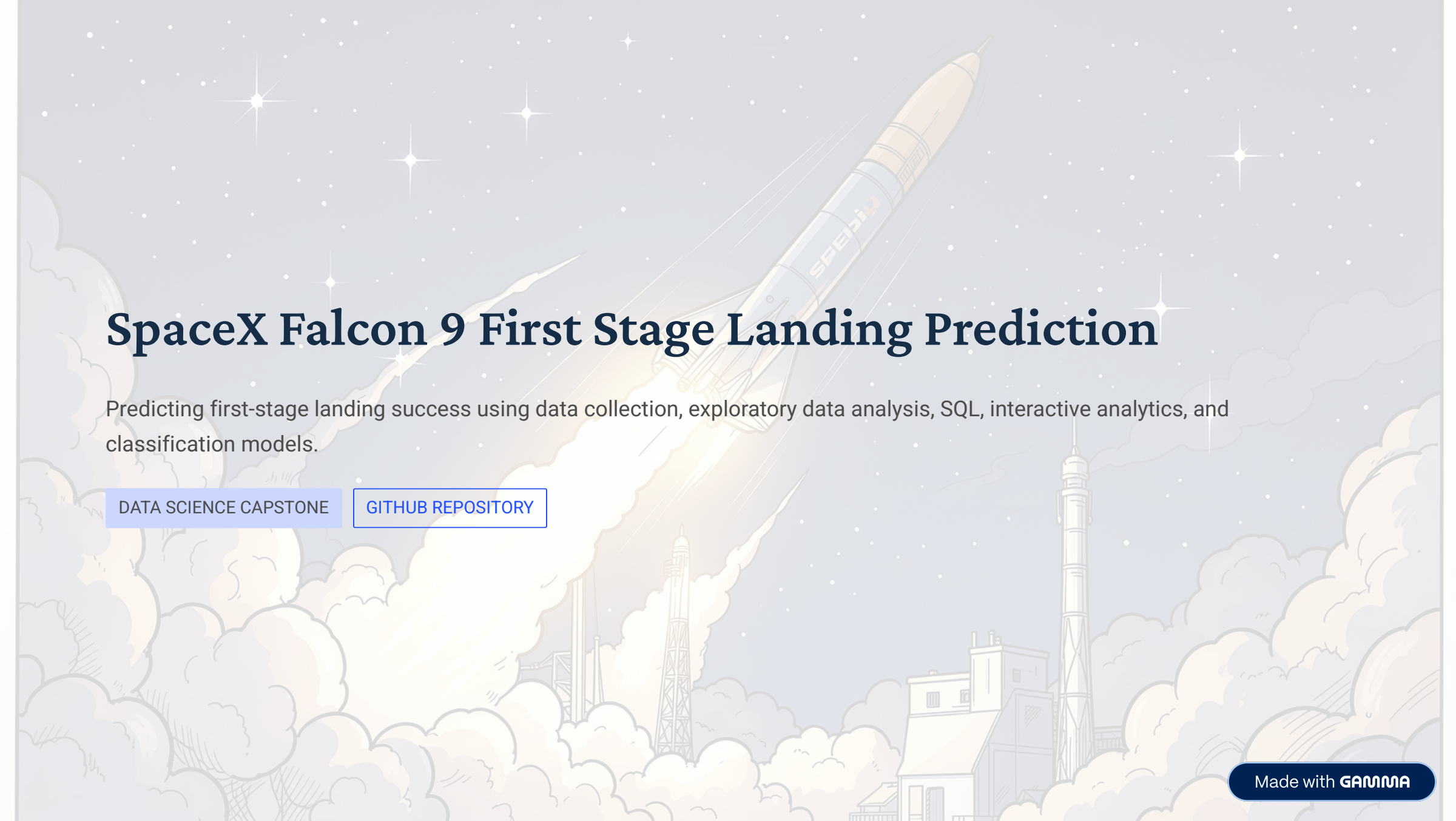


An illustration of a SpaceX Falcon 9 rocket launching from a launch pad. The rocket is shown in profile, ascending diagonally towards the top right. It has a blue body with 'SPACEX' written on it. A large, bright orange and yellow flame trail follows the rocket. The background is a dark blue night sky filled with white stars. In the foreground, there are stylized, fluffy white and yellow clouds. At the bottom, a launch pad structure with two tall service towers is visible, partially obscured by the clouds.

SpaceX Falcon 9 First Stage Landing Prediction

Predicting first-stage landing success using data collection, exploratory data analysis, SQL, interactive analytics, and classification models.

[DATA SCIENCE CAPSTONE](#)

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Executive Summary

This capstone project addresses a core operational question for SpaceX: **can historical launch data predict whether a Falcon 9 first stage will land successfully?** Using a cleaned dataset of 90 Falcon 9 launch records, the project applies a full data science pipeline — from web scraping and data wrangling through EDA, SQL analysis, interactive mapping, and classification modeling — to answer this question with measurable accuracy.

90

Launch Records

Cleaned Falcon 9 records with 18 columns

66.7%

Overall Success Rate

Proportion of successful first-stage landings

83.3%

Best Test Accuracy

Decision Tree on held-out test set

87.7%

Best CV Score

10-fold cross-validation, Decision Tree

The target variable **Class** is binary: 1 indicates a successful landing, 0 indicates failure or no attempt. Feature engineering produced 83 model inputs from numerical and one-hot encoded categorical variables. Four classifiers — Logistic Regression, SVM, Decision Tree, and KNN — were evaluated using 10-fold GridSearchCV with an 80/20 train/test split. The **Decision Tree** emerged as the best model, achieving the highest cross-validation score while matching the top test accuracy. Key predictive features include launch site, orbit class, grid fin deployment, landing leg status, booster reuse, and flight history — all of which showed strong associations with landing outcomes during EDA.

Introduction

SpaceX has fundamentally disrupted the economics of spaceflight by recovering and reusing Falcon 9 first-stage boosters. A successful landing means the same hardware can fly again, dramatically reducing per-launch costs. The business question driving this project is straightforward: **can historical launch data predict landing success before a mission flies?** Answering this has direct implications for mission planning, risk assessment, and operational readiness.

The Business Problem

If the first stage lands successfully, future launches become significantly more cost-efficient. Predicting landing outcomes from historical data enables better resource allocation, risk modeling, and mission planning — directly supporting SpaceX's reusability strategy.

Project Workflow

01

Collect

Web scrape Falcon 9 launch records from Wikipedia

02

Wrangle

Clean, encode, and structure the dataset

03

Explore

Visualize patterns with EDA and SQL

04

Interact

Build Folium maps and Plotly Dash dashboard

05

Model

Train and evaluate classification models

Data Collection & Wrangling

The data pipeline began with web scraping Falcon 9 and Falcon Heavy launch tables from Wikipedia, capturing raw flight records across multiple booster versions. The scraping notebook (jupyter-labs-webscraping.ipynb) parsed HTML tables to extract flight number, date, launch site, payload, payload mass, orbit, customer, launch outcome, booster version, and booster landing status. The raw output was saved as `spacex_launch_data.csv`.

Raw Fields Captured

- Flight number and launch date
- Launch site and pad designation
- Payload name, mass, and customer
- Orbit class (LEO, GTO, ISS, VLEO)
- Booster version and serial
- Landing outcome and pad type
- Grid fins, legs, and reuse flags

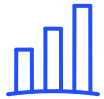
Wrangling Steps

- Converted landing outcomes to binary Class (1 = success, 0 = failure)
- Filled missing PayloadMass values with column mean
- Preserved 16 key features for modeling
- Final cleaned dataset: **90 records, 18 columns**
- Output saved as `labs-jupyter-spacex-Data wrangling results.csv`

FlightNumber	Date	BoosterVersion	PayloadMass
Integer	Date-time	Categorical	Float (kg)
Orbit	LaunchSite	Flights	Class
Categorical	Categorical	Integer	Binary target

EDA & Visual Analytics Methodology

Exploratory data analysis was conducted using `pandas`, `matplotlib`, `seaborn`, `Plotly`, `Folium`, and `Dash`. The goal was to surface patterns between landing success and operational variables — launch site, payload mass, orbit class, flight number, and year — before committing to any modeling approach. Both static and interactive visualizations were produced to support hypothesis formation and feature selection.



Static Plots

Scatter plots, bar charts, and line charts explored univariate distributions and bivariate relationships between landing outcome and key features such as payload mass, orbit, and flight number.



Dash Dashboard

A Plotly Dash application (`spacex-dash-app.py`) provided interactive controls — launch-site dropdown and payload-range slider — linked to pie charts and scatter plots for dynamic exploration of the 56-record dashboard subset.



Interactive Maps

Folium was used to plot launch sites with color-coded markers — green for successful landings, red for failures — overlaid on satellite basemaps with distance annotations to coastlines and infrastructure.



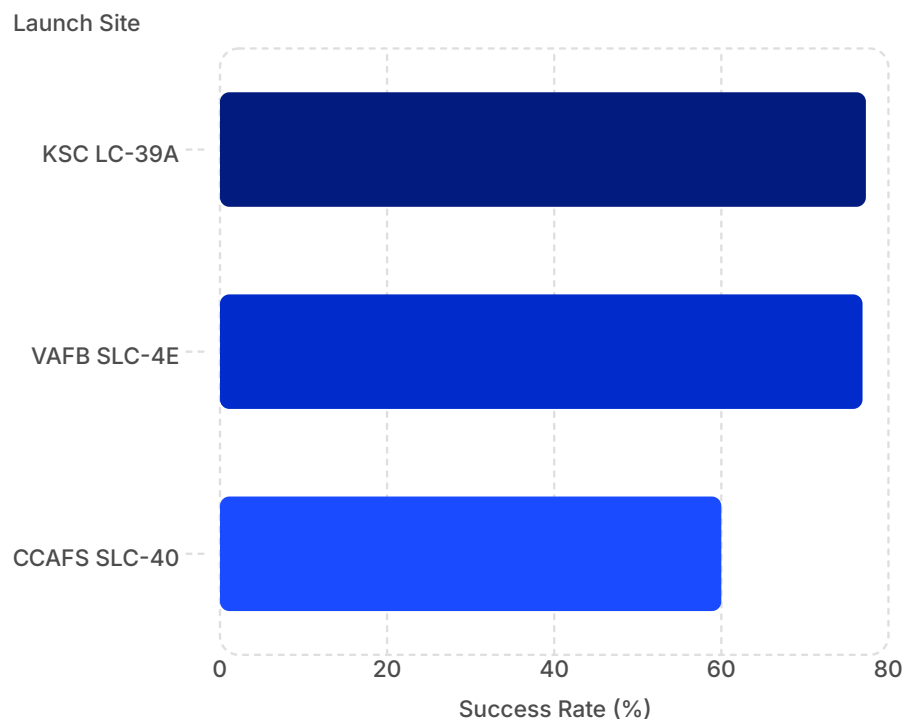
SQL Analysis

SQL queries were executed against the cleaned dataset to extract aggregated statistics: unique launch sites, total payload mass by customer, average mass by booster version, and landing outcome rankings by date range.

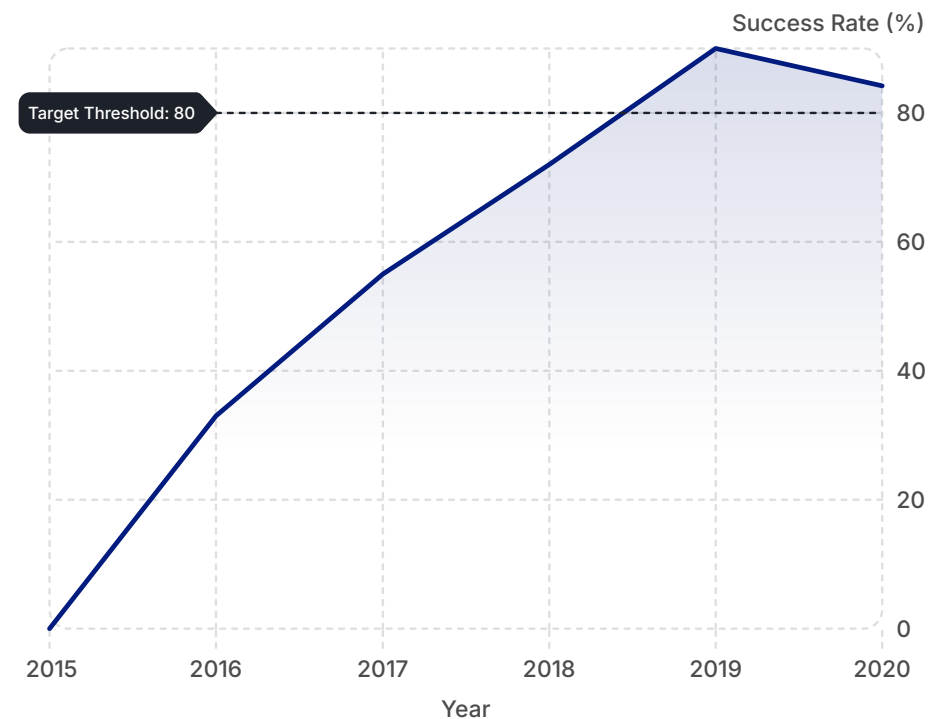
EDA Visualization Results

The EDA revealed clear, actionable patterns in landing success across time, geography, and mission configuration. Overall, **66.7% of Falcon 9 first-stage landing attempts were successful** in the cleaned dataset. Success rates varied meaningfully by launch site, with KSC LC-39A outperforming the Florida and California alternatives. The temporal trend is perhaps the most striking finding: landing reliability improved from near-zero in the earliest years to over 90% by 2019, reflecting SpaceX's rapid operational learning curve.

Launch Site Success Rates



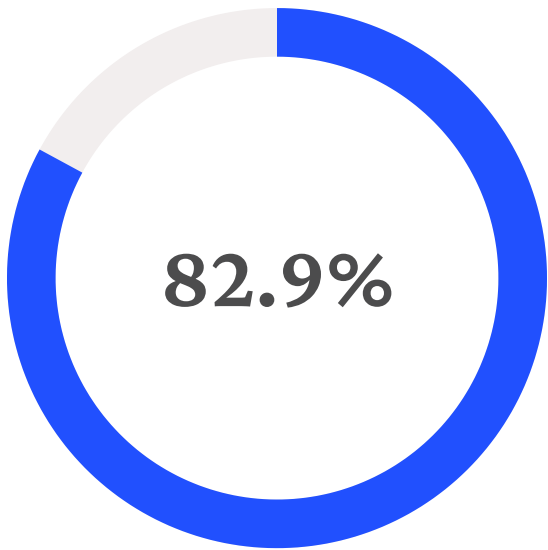
Yearly Landing Success Trend



Key Insight: Landing reliability improved substantially after 2015 as SpaceX gained operational experience. KSC LC-39A consistently outperformed other sites, likely reflecting more favorable mission profiles and infrastructure support.

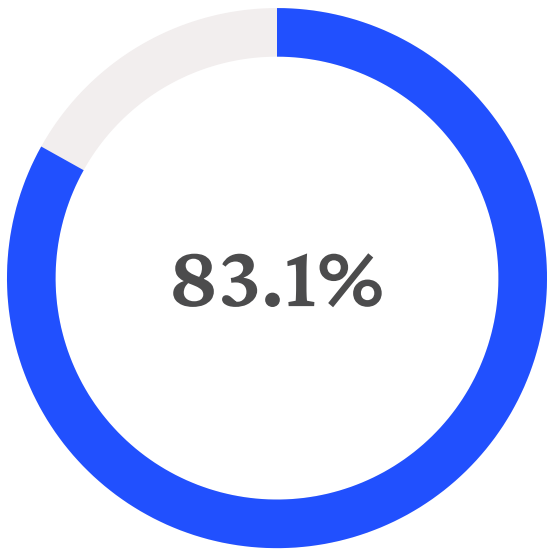
EDA Feature Insights

Beyond temporal and geographic patterns, EDA revealed that **hardware configuration and mission profile are strongly associated with landing outcome**. Grid fin deployment and landing leg status showed the largest success-rate differentials — both features are direct indicators of whether the booster was configured for a recovery attempt. Reuse status and orbit class also carried meaningful signal, suggesting that operational maturity and mission complexity both influence landing probability.



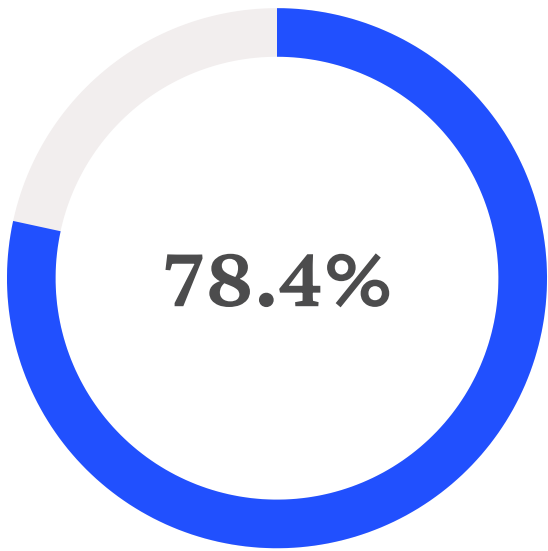
Grid Fins: Deployed

Success rate when grid fins = True vs. 10.0% when False



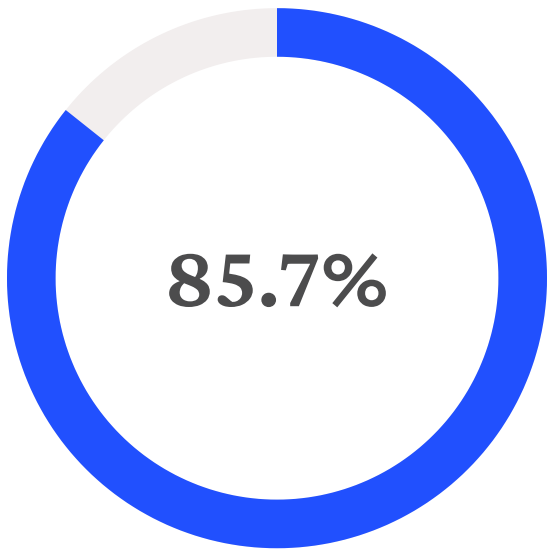
Landing Legs: Deployed

Success rate when legs = True vs. 5.3% when False



Reused Boosters

Success rate for reused vs. 58.5% for non-reused boosters



VLEO Orbit

Highest orbit-class success rate; GTO lowest at 51.9%

Orbit Class	Success Rate	Feature Type	Interpretation
VLEO	85.7%	Categorical	Lowest-energy orbit, easiest recovery
LEO	71.4%	Categorical	Most common orbit class
ISS	61.9%	Categorical	Moderate complexity
GTO	51.9%	Categorical	High-energy, hardest recovery

SQL Analysis & Interactive Map Results

SQL queries against the cleaned dataset provided aggregated operational statistics that complemented the visual EDA. Key findings include four unique launch sites in the dataset, a total NASA CRS payload mass of 45,596 kg, and an average F9 v1.1 payload mass of 2,928.4 kg. The first successful ground-pad landing occurred on **2015-12-22**. Landing outcome counts between 2010-06-04 and 2017-03-20 show that early missions frequently had no landing attempt, while successful drone-ship and ground-pad landings emerged progressively.

SQL Query Result	Value	Context
Unique Launch Sites	4	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
NASA CRS Total Payload Mass	45,596 kg	Sum across all CRS missions
Avg Payload Mass (F9 v1.1)	2,928.4 kg	Mean for F9 v1.1 booster version
First Ground-Pad Landing	2015-12-22	Historical milestone
No Attempt (2010–2017)	10	Most frequent early outcome
Success Drone Ship	5	Emerging capability
Success Ground Pad	3	Early RTLS successes

Folium Interactive Map

Launch sites were plotted with Folium markers on a satellite basemap. **Green markers** represented successful landings; **red markers** represented failures. Florida sites (KSC LC-39A, CCAFS SLC-40) cluster near Kennedy Space Center and Cape Canaveral; VAFB SLC-4E sits on the California coast. Distance markers confirmed all sites are positioned near coastlines, supporting launch safety and open downrange trajectories.

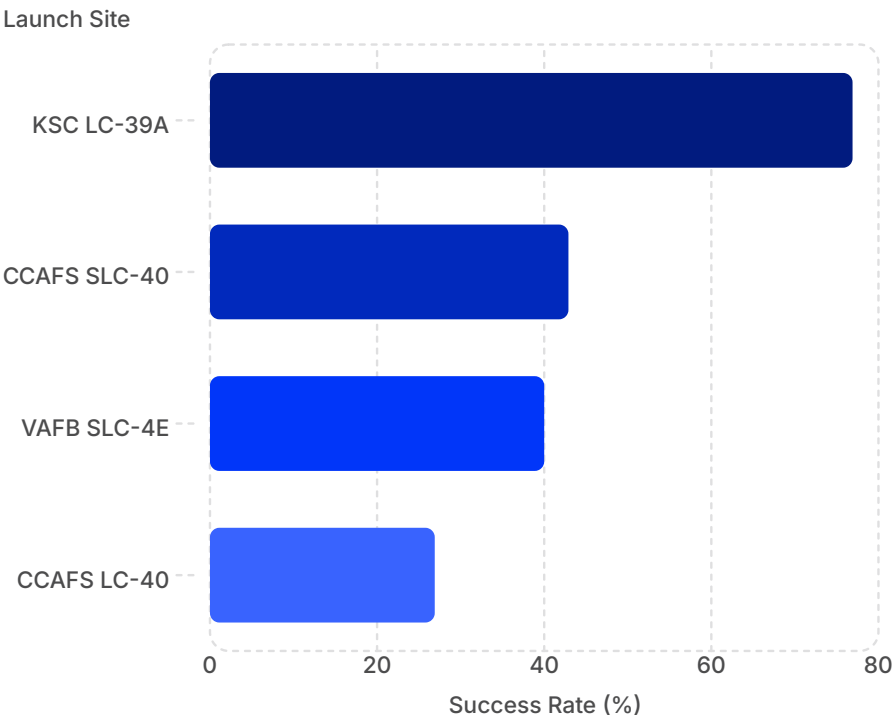
Map Features

- Color-coded success/failure markers
- Satellite basemap with terrain context
- Distance annotations to coastlines
- Proximity to highways, railways, and cities
- Popup details on marker click

Plotly Dash Dashboard Results

The Plotly Dash application (spacex-dash-app.py) provides an interactive interface for exploring landing success across launch sites and payload mass ranges. The dashboard uses a 56-record subset of the cleaned data and includes two user controls: a **launch-site dropdown** and a **payload-range slider**. Outputs include a pie chart of success/failure counts by site and a scatter plot of payload mass versus landing outcome. The dashboard makes site-level performance comparisons immediately accessible without requiring code execution.

Dashboard Site Success Rates (56 Records)



Dashboard Controls & Visuals

Launch-Site Dropdown

Filters all visuals to a single selected site for focused analysis

Payload Range Slider

Filters scatter plot by payload mass to explore mass-outcome relationships

Pie Chart

Shows success/failure proportion for the selected site and payload range

Scatter Plot

Payload mass (x-axis) vs. landing outcome (y-axis) with color encoding

Dashboard Insight: KSC LC-39A had the highest success rate (76.9%) in the dashboard subset, while CCAFS LC-40 had the lowest (26.9%). This reflects differing mission profiles across sites in the filtered dataset.

Predictive Analysis & Conclusion

Four classification models were trained and evaluated using an 80/20 train/test split (random_state=2) and 10-fold GridSearchCV for hyperparameter tuning. Features were standardized with StandardScaler, and 83 engineered inputs were created from numerical variables and one-hot encoded categoricals. All four models achieved identical test accuracy of **83.3%**, but the **Decision Tree** was selected as the best model due to its highest cross-validation score of **87.7%**.

Model	Test Accuracy	Best CV Score	Selection Status
Decision Tree	0.833	0.877	✓ Best Model
K Nearest Neighbors	0.833	0.848	Runner-up
Support Vector Machine	0.833	0.848	Runner-up
Logistic Regression	0.833	0.846	Baseline

Innovative Insights

- Landing success is not solely a payload-mass problem — grid fins and landing legs are highly predictive
- Reused boosters outperformed non-reused boosters, suggesting operational learning and reuse maturity
- The strongest improvement is temporal: success rates rose substantially after 2015
- Launch-site performance differs, indicating site-specific mission profiles matter

Final Takeaways

- Project covered full pipeline: collect, clean, explore, visualize, map, dashboard, and model
- EDA confirmed success improved over time and varied by site, orbit, and booster configuration
- Interactive Folium and Dash tools made launch-site and payload relationships explorable
- Classification models achieved **83.3% test accuracy** on Falcon 9 landing prediction
- **Decision Tree** is the recommended model for deployment



Conclusion: Falcon 9 landing success can be predicted with meaningful accuracy using historical launch, payload, site, orbit, and booster features — validating the business case for data-driven mission planning.